Causal discovery with parameter analysis

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Abstract

abstract

1 Introduction

2 The main algorithm

There given a sample set of interventions

$$V_{\mathrm{do}} = \{\mathrm{do}(V = v_0), \mathrm{do}(V = v_1), \dots\},$$

where v_i (is a) set of all nodes.

1. Assume we have samples of V_{do} (with «a» distribution) and the variables for any $V \in v$
$$S_1^v, \dots S_n^v.$$
2. For any $y \in V$, compute $\mathrm{Anc}(y)$ by:
$$\mathrm{Anc}(y) = \{v, y \not\perp\!\!\!\perp V_{\mathrm{do}}\}.$$
3. For each, $\forall\, Y \in V, \forall\, x \in \mathrm{Anc}(y)$:
   (a) estimate $w_i^{yx} = \arg\max_w P(y \mid X_{\mathrm{do}}; w)$, $1 \le i \le n$ and obtain $\hat{w}_1^{yx}, \dots, \hat{w}_n^{yx}$, where $P(y \mid X_{\mathrm{do}}; w)$ in the probability of Y from $\mathrm{Anc}(Y)$ based on all samples, assessed with X;
   (b) compute $\hat{\boldsymbol{\mu}}^{yx}$ and $\hat{\boldsymbol{\Sigma}}^{yx}$ from $\hat{w}_1^{yx}, \dots, \hat{w}_n^{yx}$;
   (c) compute $D_{\mathrm{KL}}\Big(p(w; \hat{\boldsymbol{\mu}}, \hat{\boldsymbol{\Sigma}}) \,\Big\|\, p(w; \hat{\boldsymbol{\mu}}^{xy}, \hat{\boldsymbol{\Sigma}}^{xy})\Big)$;
   (d) if $D_{\mathrm{KL}}(\,\|\,) \ge 0$, $x \in p_a(Y)$.

The first variant

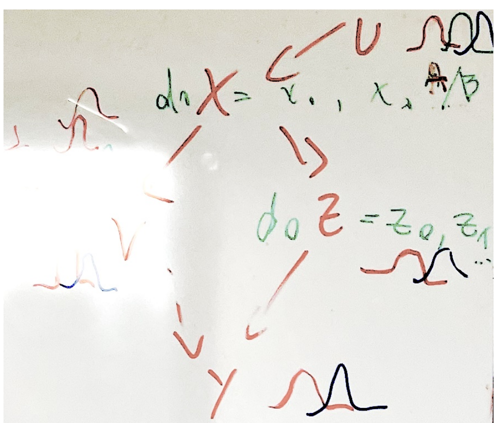
$$X_{\text{do}}\{\text{do}(X = x_0), \text{do}(X = x_1), \dots\};$$

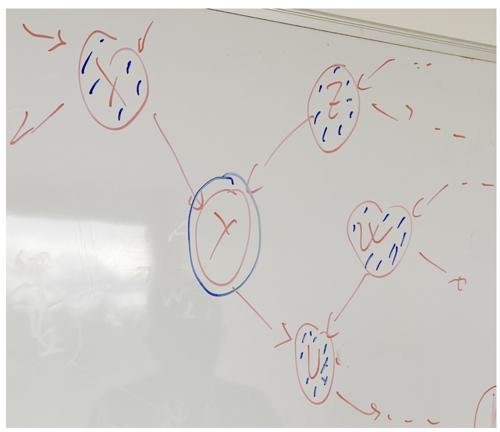
V_i set of all nodes
(a) If $X_{\text{do}} \not\perp\!\!\!\perp Y$, then $X \in \text{Anc}(Y)$

- If there exists $\text{do}(Z)$, where $Z = V \setminus X$, s.t. $X \perp\!\!\!\perp Y \mid \text{do}(Z)$, then $X \notin Pa(Y)$.
- Otherwise, $X \in \text{Pa}(Y) \rightarrow$ impossible in practice

(b) With (a) we can compute the ancestors of all variables , e.q.

$$\begin{aligned}
\text{Anc}(Y) = \{V, Z, X, V\}, \text{Anc}(X) = \{V\} \quad & => \text{Pa}(X) = \{V\} \\
\text{Anc}(V) = \{X, V\}, \text{Anc}(V) = \{\emptyset\} \quad & => \text{Pa}(V) = \{\emptyset\} \\
\text{Anc}(Z) = \{X, V\}, \quad & => ?Z \perp\!\!\!\perp X \mid V
\end{aligned}$$





3 Classification as a criterion of causal disturbance

- Set a tests for the causal disturbance: X and $X = \mathbf{x}_0$.
- Train a classifier to distinguish between the parameter distibution
- The distribution of the model parameters are the features.
- Maximize the gap between two classes

Introduction

The goal of this document is to describe the target function of the causal graph.

We treat the algorithm PCGCE in the paper *Extended summary graphs (ESG)* belongs to class of the exhaustive search algorithms. So to an optimal causal graph we make no difference between

1) exhaustive printing or combinatorial algorithms,

2) gradient-based algorithms including algorithms with relaxations,

3) zero-gradient algorithms.

The only thing we expect its result is a causal graph, optimal according to this criterion.

The model structure includes:

1) the time is continuous $\frac{dx}{dt} = f(x,t)$ (basic: discrete with $\Delta t = 1$),

2) the time series X, Y, Z has a spacial structure $\frac{dx}{dx} = g(x,s)$ (basic: scalar-valued),

3) there is no limitations of the time as in the ESG,

4) the model is linear in the latent space (basic: linear in the observable space),

5) the latent space represent a spectrum of the observable domain.

The graph structure:

1) the graph is a bipartite graph between the observable and the latent,

2) the direct graph is isomorphic to the reverse graph,

3) the causal graph is in the hypergraph that groups the variables or the time.

The time structure for the convolutional model $\int_{\tau=0}^{T} f(\tau)h(t-\tau)d\tau$

The causal graph $G = G(t)$ is a time series. We suppose either:

1) it is stationary fot all t,

2) it is stationary for $t \geq T$ for with the period T,

3) it is non-stationary but with rejection to discover its structure.

4 Appendix. Natural Gradient Descent

The natural gradient descent is a method to optimize the model parameters θ by estimating the distribution

$$p(\theta) \propto \exp(-\mathcal{L}(\theta))$$

where $\mathcal{L}(\theta)$ is the loss function of the model. The natural gradient descent is defined as

$$\theta_{t+1} = \theta_t - \eta F^{-1}$$

where F is the Fisher information matrix of the model parameters θ.

Fisher Metric and Natural Gradient

When working with probabilistic models, we can use a metric that is more natural for probability distributions: the Kullback–Leibler (KL) divergence. Suppose we have a likelihood model

$$p(x \mid \theta),$$

where x denotes fixed data and θ is a parameter vector.

Let θ_t denote the current parameter values, and suppose we wish to find updated parameters θ_{t+1}. Under the natural gradient framework, this is done by solving the following optimization problem:

$$\theta_{t+1} = \arg\min_{\theta} \left[f(\theta_t) + \nabla f(\theta_t)^\top (\theta - \theta_t) + D_{\mathrm{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t)) \right].$$

In general, this minimization problem is intractable. However, the KL divergence can be approximated locally using a second-order Taylor expansion. This expansion yields the Fisher information matrix F, which captures the local curvature of the statistical manifold. In particular, around θ_t, we have

$$D_{\mathrm{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t)) \;\approx\; \frac{1}{2}(\theta - \theta_t)^\top F (\theta - \theta_t),$$

where the Fisher information matrix is defined as

$$F = \mathbb{E}_{x \sim p(x|\theta)} \left[\big(\nabla_\theta \log p(x \mid \theta)\big) \big(\nabla_\theta \log p(x \mid \theta)\big)^\top \right].$$

The Fisher information matrix therefore encodes the curvature of the likelihood-based loss function in parameter space. Using this approximation, the resulting natural gradient update rule becomes

$$\theta_{t+1} = \theta_t - \gamma F^{-1} \nabla f(\theta_t),$$

where $\gamma > 0$ is the learning rate.

Natural Gradient from an Information-Geometric Perspective

Consider a parametric family of probability distributions

$$\mathcal{M} = \{ p(x \mid \theta) \mid \theta \in \Theta \subset \mathbb{R}^d \},$$

which forms a smooth statistical manifold $\mathcal{M}$. Each parameter vector θ corresponds to a point on this manifold, and the coordinate system θ is generally non-Euclidean.

The natural notion of distance between nearby probability distributions on $\mathcal{M}$ is given by the Kullback–Leibler divergence. For two infinitesimally close distributions $p(x \mid \theta)$ and $p(x \mid \theta + d\theta)$, the KL divergence admits the second-order expansion

$$D_{\mathrm{KL}}(p(x \mid \theta + d\theta) \parallel p(x \mid \theta)) = \frac{1}{2} d\theta^\top G(\theta) d\theta + o(\|d\theta\|^2),$$

where

$$G(\theta) = \mathbb{E}_{x \sim p(x|\theta)} \left[\big(\nabla_\theta \log p(x \mid \theta) \big) \big(\nabla_\theta \log p(x \mid \theta) \big)^\top \right]$$

is the **Fisher information metric tensor** on $\mathcal{M}$.

Let $f(\theta)$ be a smooth objective function defined on the manifold (e.g., the negative log-likelihood or empirical risk). Given the current point $\theta_t \in \mathcal{M}$, we seek a new point θ_{t+1} by minimizing a first-order approximation of f subject to a constraint on the geodesic distance induced by the Fisher metric. This leads to the optimization problem

$$\theta_{t+1} = \arg\min_\theta \left\{ f(\theta_t) + \nabla f(\theta_t)^\top (\theta - \theta_t) \;\middle|\; D_{\mathrm{KL}}(p(x \mid \theta) \parallel p(x \mid \theta_t)) \le \epsilon \right\}.$$

Using the local quadratic approximation of the KL divergence, this constrained problem reduces to minimizing the linearized objective under a Riemannian norm constraint. The solution is given by a step in the direction of the **natural gradient**, defined as

$$\widetilde{\nabla} f(\theta) = G(\theta)^{-1} \nabla f(\theta).$$

Accordingly, the natural gradient descent update takes the form

$$\theta_{t+1} = \theta_t - \gamma \widetilde{\nabla} f(\theta_t) = \theta_t - \gamma G(\theta_t)^{-1} \nabla f(\theta_t),$$

where $\gamma > 0$ is the step size.

From an information-geometric standpoint, this update corresponds to moving along the direction of steepest descent with respect to the Fisher–Rao Riemannian metric, ensuring invariance under smooth reparameterizations of the statistical manifold.

5 Second document

Introduction

We construct a Bayesian causal-inference model for multivariate time series. The goal is to identify a causal relationship between time series given the constructed model. We suppose that the unknown dynamics generated the observable time series from the latent space. Here we keep the model simple to expand it for 1) multimodal time series and control problems, 2) transformers and large models, 3) continuous-time dynamical systems and Neural ODE.

6 Discrete-time linear model

There given time series $\mathbf{x}_t$ as observations. The time series are generated by a discrete-time dynamical system in the latent space $\mathbf{z}_t$ with unknown dynamics:

$$\mathbf{x}_{t+1} = f\big(\mathbf{x}_t\big) + \boldsymbol{\zeta}_{t+1}, \tag{1}$$

where $\boldsymbol{\zeta}_t$ is the noise term. The linear model is

$$\mathbf{x}_t = \mathbf{W}\mathbf{x}_{t-1} + \mathbf{b} + \boldsymbol{\zeta}_t.$$

The time series are represented via *delay embedding*. The matrix of delay vectors for the scalar-valued series x_t is the Hankel matrix $\mathbf{X}$. Let the vector $\mathbf{x}_{t-1}$ be the first column of the matrix $\mathbf{X}_{t-1}$. Each column of the matrix X contains a point $\mathbf{x}_t$. Call *the phase trajectory* of a dynamical system the sequence $\{\mathbf{x}_t\}$. The dimensionality of the model is

$$\begin{pmatrix} w_{1,1} & w_{1,2} & \cdots & w_{1,N} \\ w_{2,1} & w_{2,2} & \cdots & w_{2,N} \\ \vdots & \vdots & w_{\kappa,\tau} & \vdots \\ w_{K,1} & w_{K,2} & \cdots & w_{K,N} \end{pmatrix} \underbrace{\begin{pmatrix} x_{t-1} & x_{t-2} & \cdots & x_{t-(N+T)} \\ x_{t-2} & x_{t-3} & \cdots & x_{t-(N+T-1)} \\ \vdots & \vdots & x_{t-\tau} & \vdots \\ x_{t-N} & x_{t-(N-1)} & \cdots & x_T \end{pmatrix}}_{\text{Hankel matrix } \mathbf{X}_{t-1}}.$$

The model parameters are

$$w_{\kappa,\tau},\ \tau = 1, \dots, K,\ \kappa = 1, \dots, N.$$

For the vector autoregression model $N = K$. Otherise suppose $N \gg K$ and call N the original embedding dimension and K the reduced embedding dimension. The discrete time lag $\Delta t = 1$ and $T \ni \{1, \dots, t\}$ is the sample size. For d-variate time serires $\mathbf{x}_t^{(1,\dots,d)}$, the matrix $\mathbf{X}$ is block-Hankel with d blocks. The matrix $\mathbf{W}$ in this case is $(K \times d \cdot N)$.

7 State space reconstruction

Introduce the singular value decomposition of the linear operator in the model (1):

$$f = \mathbf{W}\mathbf{\Lambda}\mathbf{V}^\mathsf{T}.$$

Denote by $\mathbf{z}$ the variable in the latent space. It corresponds to the dynamics of the system,

$$\begin{tikzcd}
\mathbf{x} \arrow[rr, "f"] \arrow[dr, "\mathbf{W}"'] & & \mathbf{y} \\
& \mathbf{z} \arrow[ur, "\mathbf{\Lambda}\mathbf{V}^\mathsf{T}"'] & \\
& t \arrow[u, "p(\mathbf{z}|t)"] &
\end{tikzcd}$$

For simplicity omit the time t and the bias $\mathbf{b}$ in the model (1). Let the variable y be the forecast of the time series x_t. The variable $\mathbf{z}$ is the projection of the observable $\mathbf{x}$ onto the low-dimensional latent space.

In the terms of neural networks, this superposition is an encoder and a decoder. The linear operators $\mathbf{W}$ and $\mathbf{V}$ are the orthogonal and $\mathbf{\Lambda}$ is diagonal with positive descending elements. Each row of the matrix W

$$\mathbf{W}\mathbf{x} = \begin{pmatrix} \mathbf{w}_1^\mathsf{T} \\ \mathbf{w}_2^\mathsf{T} \\ \vdots \\ \mathbf{w}_K^\mathsf{T} \end{pmatrix} \mathbf{x}$$

produces an element of a vector $\mathbf{z}$ in the latent space by the dot product

$$z_i = \mathbf{w}_i^\mathsf{T}\mathbf{x}.$$

Call the *embedding* the linear map $\mathbf{z} = \mathbf{W}\mathbf{x}$.

The system dynamics is represented by the generative model $p(\mathbf{z} \mid t)$. It approximates the embedding in the latent space with the support t.

8 Maximum a posteriori estimation of the parameters

The posterior distribution of the model parameters $\mathbf{w}$ given the data $\{\mathbf{x}, \mathbf{y}\}$ is

$$p(\mathbf{w}|\mathbf{x}, \mathbf{y}) = \frac{p(\mathbf{y}|\mathbf{x}; \mathbf{w})p(\mathbf{w}|\mathbf{x})}{p(\mathbf{y}|\mathbf{x})}.$$

The likelihood function is

$$p(\mathbf{y}|\mathbf{x}; \mathbf{w}) = \prod_{t=1}^{T} p(\mathbf{y}_t|\mathbf{x}_t; \mathbf{w}).$$

The error given the posterior distribution of the parameters is

$$\mathcal{L}(\mathbf{w}) = \frac{1}{2}\underbrace{(\mathbf{f}-\mathbf{y})^\mathsf{T}\mathbf{B}^{-1}(\mathbf{f}-\mathbf{y})}_{\text{reconstruction error}} + \frac{1}{2}\underbrace{(\mathbf{w}-\boldsymbol{\mu})^\mathsf{T}\mathbf{A}^{-1}(\mathbf{w}-\boldsymbol{\mu})}_{\text{error in parameters}}.$$

Let the matrix $\mathbf{B} = \beta\mathbf{I}$ for the hypothesis that the variance β of the reconstruction error the same in the time segment $(1,\dots,T)$. The variance β is estimated as empirical distribution from the data over time t. The covariance $\mathbf{A}$ and the mean $\boldsymbol{\mu}$ of the normally-distributed parameters are estimated as empirical distribution from the data.

9 The criterion of causality

The criterion of the causal relationship between the elements of the model, we use the Kullback-Leibler divergence between the distributions of the parameters given the intervention $\mathrm{do}(\mathbf{x})$:

$$\mathrm{KL}\big(p(\mathbf{w}|\mathbf{x},\mathbf{y}, \,||\, p(\mathbf{w}|\mathrm{do}(X=\mathbf{x}_0),\mathbf{y}\big),$$

where $\{\mathbf{w}\} = \mathbf{W}$ is the set of the model parameters. The parameters are maximized according to the maximum a posteriori criterion.

For a linear operator $\mathbf{W}$, elimination of an edge from a node X, an element of the vector $\mathbf{x}_t$, is equivalent to zeroing the corresponding item $w_{\kappa,\tau}$. The node X corresponds to a column in $\mathbf{W}$. Since the parameter w is a random variable, the following statement holds, see Table 9.

Table 1: The action for a node based on the variance of the parameter w.

Expectation	**Variance**	**Statement**	**Action**
0	tends to 0	Sure that there is no impact	Remove the node
0	not tends to 0	Uncertain about the impact	Research to remove
not 0	tends to 0	Sure that there is an impact	Assert the node
not 0	not tends to 0	The impact must be specified	Research to assert
any	large values	Sure that there is no impact	Remove the node

10 The intervention algorithm

1. One step of back-prop in one layer.
2. Compute the covariance of the parameters (note the relation between Cov, Hessian and Fisher matrices).
3. Select a node X to make an intervention.
4. Make an intervention $\mathrm{do}(X = \mathbf{x}_0)$ for a given node.

5. Compute the KL divergence between the distributions of the parameters with and without intervention.

6. Remove the intervention and go to step 1 until all nodes are processed.

7. The causal graph is constructed as a directed graph with edges weighted by the KL divergences.

11 do(X) intervention policy

For regression or one-class classification, $do(X = x)$ is considered with values of x that lie outside the distribution of the time series at time t. For two-class classification, $do(X = x)$ is considered with values of x equal to the expectation of the opposite class.

What happens after the intervention? Until the forecasting and reconstruction error remains the same:

$$\mathcal{L}\big(p(\mathbf{y}|\mathbf{x};\mathbf{w})\big) = \text{const},$$

1. Until the end of the short-term forecast horizon.

2. From one of sides, either X or Z, we start the intervention.

3. Start with the parameter of maximum variance of reconstruction error

4. Then, move to the parameters with lower variance.

5. Estimate the KL divergence of maximum posterior after each intervention.

6. Analyse the expectation and the variance of the parameters.

7. Eliminate the parameters with the highest KL divergence.

8. Keep the parameters with the lowest KL divergence.

9. Store them in the graph of causality.

Compare two graphs: direct and reverse.

12 Sequential causal inference

Results of the computational experiment: the norm of the gradient $\|\frac{\partial\mathcal{L}}{\partial\mathbf{W}}\|_2$ of the layer (l) causes the norm of the gradient in the previous layer $l-1$

$$\left\|\frac{\partial\mathcal{L}}{\partial\mathbf{W}^{(l)}}\right\|_2 \rightsquigarrow \left\|\frac{\partial\mathcal{L}}{\partial\mathbf{W}^{(l-1)}}\right\|_2.$$

As well as the error on the training set causes the error on the validation set,

$$L_{\text{train}} \rightsquigarrow L_{\text{validation}}.$$

This holds for different optimization sessions.

The backprop procedure compute the parameters and their variance for sequence of layers. For several layers of a neural network

$$f = \boldsymbol{\sigma}^{(L)} \circ \mathbf{W}^{(L)} \circ \boldsymbol{\sigma}^{(l-1)} \circ \mathbf{W}^{(l-1)} \circ \cdots \circ \boldsymbol{\sigma}^{(1)} \circ \mathbf{W}^{(1)}\mathbf{x}.$$

13 The Hessain, Fisher, and covariance matrices

The Hessian matrix of the loss function $\mathcal{L}$ is defined as

$$\mathbf{H} = \frac{\partial^2 \mathcal{L}}{\partial \mathbf{w} \partial \mathbf{w}^\mathsf{T}}.$$

Using the posterior of the parameters, we have

$$\mathbf{H} = \mathbb{E}\left[\frac{\partial^2 \log p(\mathbf{w}|\mathbf{x}, \mathbf{y})}{\partial \mathbf{w} \partial \mathbf{w}^\mathsf{T}}\right].$$

The Fisher information matrix is defined as

$$\mathbf{F} = \mathbb{E}\left[\Big(\frac{\partial \log p(\mathbf{w}|\mathbf{x}, \mathbf{y})}{\partial \mathbf{w}}\Big)\Big(\frac{\partial \log p(\mathbf{w}|\mathbf{x}, \mathbf{y})}{\partial \mathbf{w}}\Big)^\mathsf{T}\right].$$

Under certain regularity conditions, the following relation holds:

$$\mathbf{H} = -\mathbf{F}.$$

The covariance matrix of the parameters $\mathbf{w}$ is defined as

$$\mathbf{A} = \mathbb{E}\Big[(\mathbf{w} - \mathbb{E}[\mathbf{w}])(\mathbf{w} - \mathbb{E}[\mathbf{w}])^\mathsf{T}\Big].$$

Under certain assumptions, the following relation holds:

$$\mathbf{A} = \mathbf{F}^{-1}.$$

Causal intervention in the backpropagation procedure

The fully connected feedforward neural network with L layers $\ell = 1, \dots, L$ has the weight matrix $\mathbf{W}^{(\ell)} \in \mathbb{R}^{n_\ell \times n_{\ell-1}}$, the bias vector $\mathbf{b}^{(\ell)} \in \mathbb{R}^{n_\ell}$, pre-activation $\mathbf{z}^{(\ell)} \in \mathbb{R}^{n_\ell}$, activation $\mathbf{a}^{(\ell)} \in \mathbb{R}^{n_\ell}$ and $\mathbf{a}^{(0)} = \mathbf{x}$ is the input.

The forward propagation, for $\ell = 1, \dots, L$:

$$\mathbf{z}^{(\ell)} = \mathbf{W}^{(\ell)}\mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)} \tag{2}$$

$$\mathbf{a}^{(\ell)} = \boldsymbol{\sigma}^{(\ell)}\big(\mathbf{z}^{(\ell)}\big) \tag{3}$$

where $\boldsymbol{\sigma}^{(\ell)}(\cdot)$ is the activation function of layer ℓ. The network output is $\hat{\mathbf{y}} = \mathbf{a}^{(L)}$.

The loss function $\mathcal{L}(\hat{\mathbf{y}}, \mathbf{y})$ is a scalar loss function comparing prediction $\hat{\mathbf{y}}$ with target $\mathbf{y}$.

Backpropagation. Define the error signal (gradient of the loss with respect to pre-activation):

$$\boldsymbol{\delta}^{(\ell)} \equiv \frac{\partial \mathcal{L}}{\partial \mathbf{z}^{(\ell)}}$$

Output Layer. Using the chain rule:

$$\boldsymbol{\delta}^{(L)} = \frac{\partial \mathcal{L}}{\partial \mathbf{a}^{(L)}} \odot \boldsymbol{\sigma}'^{(L)}\left(\mathbf{z}^{(L)}\right), \tag{4}$$

where $\odot$ denotes element-wise multiplication.

Hidden Layers. For $\ell = L-1, \ldots, 1$:

$$\boldsymbol{\delta}^{(\ell)} = \left(\mathbf{W}^{(\ell+1)}\right)^{\top} \boldsymbol{\delta}^{(\ell+1)} \odot \boldsymbol{\sigma}'^{(\ell)}\left(\mathbf{z}^{(\ell)}\right) \tag{5}$$

Gradients with Respect to Parameters

Using the definition of $\boldsymbol{\delta}^{(\ell)}$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(\ell)}} = \boldsymbol{\delta}^{(\ell)} \left(\mathbf{a}^{(\ell-1)}\right)^{\top} \tag{6}$$

$$\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(\ell)}} = \boldsymbol{\delta}^{(\ell)} \tag{7}$$

Vectorized Form.

For each layer ℓ:

$$\begin{aligned}
\mathbf{z}^{(\ell)} &= \mathbf{W}^{(\ell)}\mathbf{a}^{(\ell-1)} + \mathbf{b}^{(\ell)} \\
\mathbf{a}^{(\ell)} &= \boldsymbol{\sigma}^{(\ell)}(\mathbf{z}^{(\ell)}) \\
\boldsymbol{\delta}^{(\ell)} &= \begin{cases} \dfrac{\partial \mathcal{L}}{\partial \mathbf{a}^{(L)}} \odot \boldsymbol{\sigma}'^{(L)}(\mathbf{z}^{(L)}), & \ell = L \\ \left(\mathbf{W}^{(\ell+1)}\right)^{\top} \boldsymbol{\delta}^{(\ell+1)} \odot \boldsymbol{\sigma}'^{(\ell)}(\mathbf{z}^{(\ell)}), & \ell < L \end{cases} \\
\frac{\partial \mathcal{L}}{\partial \mathbf{W}^{(\ell)}} &= \boldsymbol{\delta}^{(\ell)}(\mathbf{a}^{(\ell-1)})^{\top} \\
\frac{\partial \mathcal{L}}{\partial \mathbf{b}^{(\ell)}} &= \boldsymbol{\delta}^{(\ell)}
\end{aligned}$$

14 Computational experiment

For control problem we investigate a driven pendulum problem. A pendulum of a given length has its own period of oscillation. Driven by an external force a pendulum had forced oscillation. These two oscillations are simple and independent. They superposition are observed. A change in the frequency of the driving force disturbs the superposition. The goal is to identify the causal relationship between the driving force and the normal mode of the pendulum oscillations.

15 Appendix: Miscellaneous

Todo: Put plate notations for the bayesian model structure Γ in the slides (on monday). Introduce the bayesian structure of model Γ and associate it with the causal graph G. (The distribution ofΓ could be coincided with the weights of the edges in G.) The weights of the arrows in the causation model means the posterior model structure Γ.

In the section 1: construct the manifold from the set of samples $\{\mathbf{z}\}$. Then control this manifold and the variables $do(Z = \mathbf{z})$ to infer the forecasting model f. Weight change and distribution as new data. Change parameters of the generative model $p(\mathbf{z}|t)$ and observe changes in the forecast. Use Variational Autoencoder instead of Normalising Flows. It simplifies the generative model.

Put the idea how to compute CI for continuous time dynamical system.

Discussion in January. Check the error: Infer another time series. It changes parameter, but does not changes the forecast. The error can stay the same.

Insert into section 2: Intervention in neuron weights versus intervention in the input data: proof of pros and cons.

We apply do() the sin to reconstruct the trajectory. We know the sin since we know the trajectory could be reduced to dim=2.

Review: Hypothesis of *mélange* by M.S. Pinsker, or Ergodic Theory by Y.G. Sinai is exactly the Causal Inference.

Afterlife: When a Neural Network is a solution of the differential equation? Use the Neural ODE to test Causality in the mathematical models.

Domain transfer by causality graph: same model different datasets

16 Appendix: Control problem and Canonical Correlation Analysis

$$
\begin{tikzcd}
\mathbf{X} \arrow[rr, "F"] \arrow[dr, "\mathbf{U}"'] & & \mathbf{Y} \arrow[dl, "\mathbf{V}"] \\
& \mathbf{P}, \mathbf{Q} & \\
& t \arrow[u, "p(\mathbf{z})"] &
\end{tikzcd}
$$

17 Appendix: continuous time dynamical system and delay embedding

For the continuous-time dynamical system, the state vector $\mathbf{z}(t)$ evolves according to the differential equation

$$\frac{d\mathbf{z}(t)}{dt} = f\big(\mathbf{z}(t), \mathbf{x}(t), \mathbf{y}(t)\big) + \boldsymbol{\varepsilon}(t),$$

and the observed time series are generated from the state vector via the observation functions:

$$\mathbf{x}(t) = h_x\big(\mathbf{z}(t)\big) + \boldsymbol{\xi}_x(t), \quad \mathbf{y}(t) = h_y\big(\mathbf{z}(t)\big) + \boldsymbol{\xi}_y(t),$$

where $\boldsymbol{\varepsilon}(t)$, $\boldsymbol{\xi}_x(t)$, and $\boldsymbol{\xi}_y(t)$ are noise terms.

The matrix of delay vectors for the scalar-valued series x_t is the Hankel matrix:

$$\mathbf{X} = \begin{pmatrix} x(t_1) & x(t_2) & \cdots & x(t_{N-T}) \\ x(t_1 - \Delta t) & x(t_2 - \Delta t) & \cdots & x(t_{N-T} - \Delta t) \\ \vdots & \vdots & \ddots & \vdots \\ x(t_1 - (T-1)\Delta t) & x(t_2 - (T-1)\Delta t) & \cdots & x(t_{N-T} - (T-1)\Delta t) \end{pmatrix},$$

where T is the length of the time series sample, $t \in (1, T)$. The time lag is Δt and N is the embedding dimension.

The continuous-time forecasting model is a convolution of two functions: the input signal $x(t)$ and the kernel $K(t)$ that represents the system dynamics. Thus, the output signal $y(t)$ can be expressed in the kernel terms:

$$y(t) = \int_{t-T}^{t} K(t-\tau)x(\tau)d\tau + \varepsilon(t).$$

The vector form of this equation is:

$$\mathbf{y}_t = \mathbf{K}^{\mathsf{T}}\mathbf{x}_t - \boldsymbol{\zeta}_t,$$